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Università degli Studi di Bergamo

Dipartimento di Ingegneria Gestionale,
dell'Informazione e della Produzione

Deep Learning Course

Assignment 2

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Contents

1 Introduction **1**

2 Hyperparameter Tuning and Model Selection **2**
2.1 Configuration Summary 3
2.2 Learning Curves 3
2.3 Discussion 4

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1 Introduction

Objective. The goal of this assignment is to implement a hybrid ResNet-Transformer classifier for image classification on a 9-class waste classification dataset (Cardboard, Food Organics, Glass, Metal, Miscellaneous Trash, Paper, Plastic, Textile Trash, Vegetation).

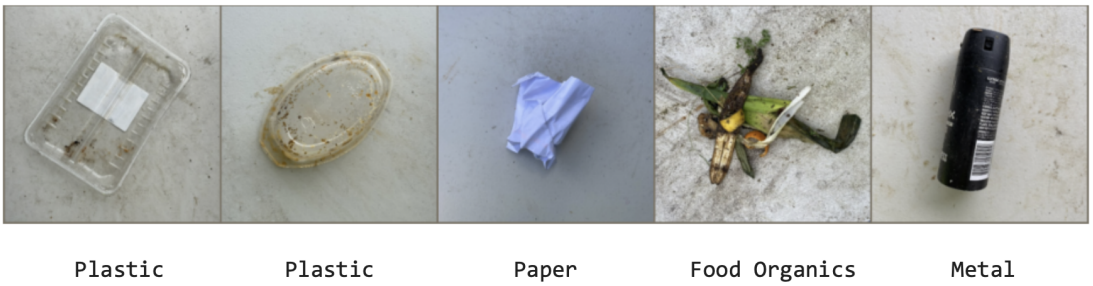


Figure 1: Sample images from the dataset with their corresponding class labels.

Architecture overview. The model combines a pre-trained ResNet-18 backbone for feature extraction with a Transformer encoder for capturing global relationships between image patches.

Table 1: ResNet-Transformer architecture components.

Component	**Description**
ResNet-18 backbone	Pre-trained on ImageNet, last two layers removed
Feature maps	$7 \times 7 \times 512$ dimensional output
Projection layer	Maps 512-dim features to embedding dimension
CLS token	Learnable token prepended for classification
Positional embeddings	Added to encode spatial information
Transformer encoder	Multi-head self-attention layers
Classification head	LayerNorm + Linear on CLS token output

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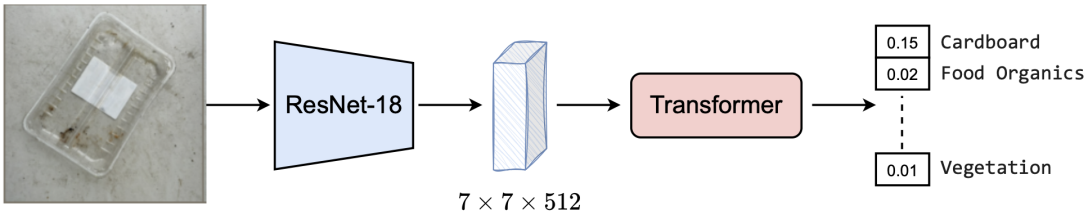


Figure 2: ResNet-Transformer architecture: the input image is processed by ResNet-18 to produce $7 \times 7 \times 512$ feature maps, which are then fed to the Transformer encoder for classification.

Training setup. An 80/20 train-validation split was used for all experiments. The ResNet-18 backbone was frozen to reduce memory usage and training time, except in Configuration 4 where the last convolutional block (layer4) was unfrozen. All models were trained with Adam optimizer (lr=10^{-4}), CrossEntropyLoss, and early stopping with patience of 5 epochs. Training was performed on a MacBook Air M3 (CPU) 16GB RAM. The best model weights are hosted on HuggingFace for evaluation (https://huggingface.co/aminb00/assignment2) and ready to use in Test Notebook.

2 Hyperparameter Tuning and Model Selection

Accuracy and loss curves for all tested configurations are provided below and in the accompanying notebook.
Explored configurations:

- **Embedding dimension**: 128, 256, 512
- **Number of attention heads**: 4, 8
- **Number of transformer layers**: 2, 3, 4
- **Data augmentation**: RandomCrop, RandomHorizontalFlip, ColorJitter
- **Backbone fine-tuning**: Unfreezing the last ResNet block (layer4)

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2.1 Configuration Summary

Table 2: Summary of all tested configurations and their validation accuracy.

Config	EMBED_DIM	HEADS	LAYERS	Extras	Best Val Acc (%)
1	256	8	2	—	78.05
2	128	4	3	—	77.14
3	512	8	4	—	79.10
4	**256**	**8**	**2**	**Aug + Unfreeze**	**80.15**

2.2 Learning Curves

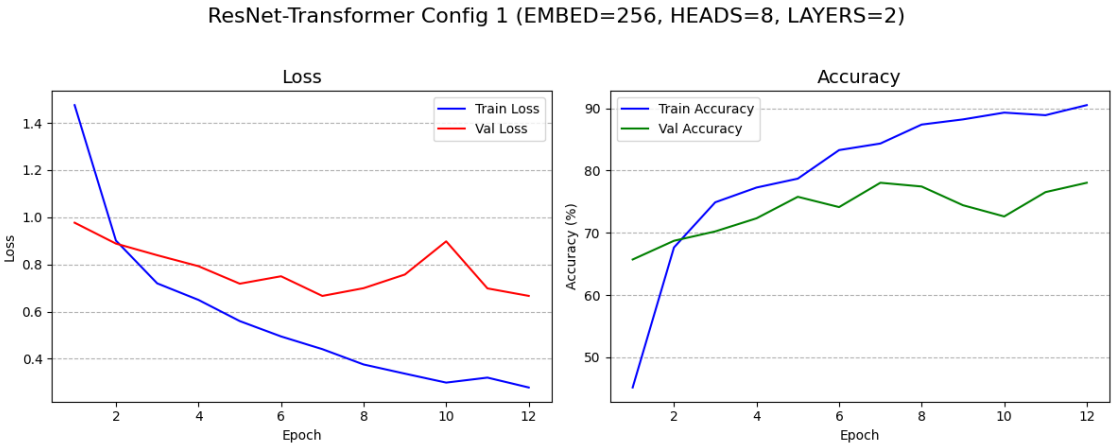


Figure 3: Config 1 (EMBED=256, HEADS=8, LAYERS=2) — Best: 78.05%, Final: 78.05%.

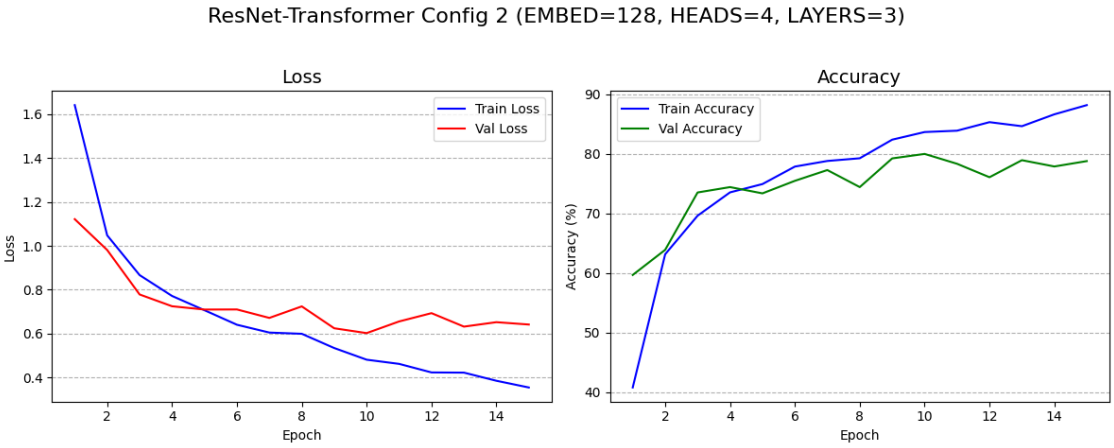


Figure 4: Configuration 2 (EMBED=128, HEADS=4, LAYERS=3) — Best: 77.14%, Final: 76.09%.

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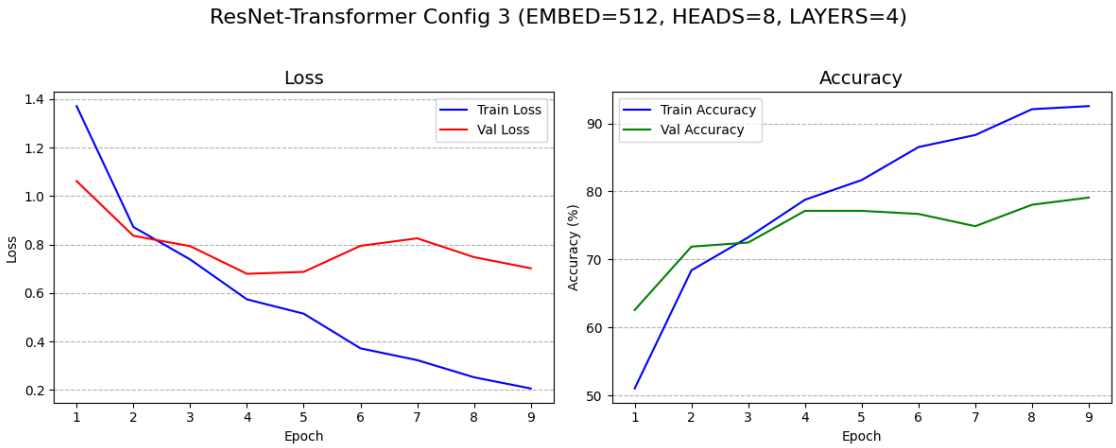


Figure 5: Configuration 3 (EMBED=512, HEADS=8, LAYERS=4) — Best: 79.10%, Final: 79.10%.

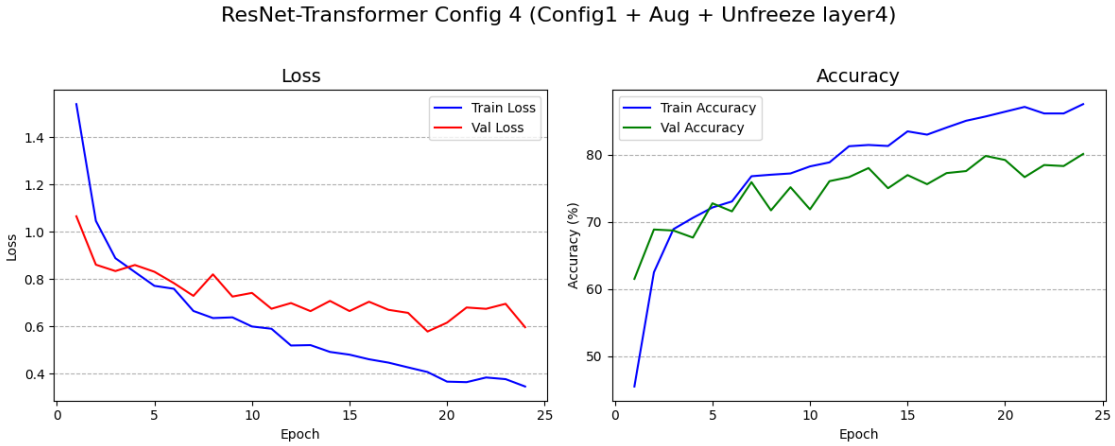


Figure 6: Configuration 4 (Config 1 + Aug + Unfreeze layer4) — Best: 80.15%, Final: 80.15%.

2.3 Discussion

Configuration 4 achieves the best validation accuracy at **80.15%**, demonstrating that data augmentation and partial backbone fine-tuning provide the most significant improvements. Reducing model capacity (Config 2: 77.14%) slightly hurts performance compared to the baseline (Config 1: 78.05%), while increasing capacity with a larger embedding dimension and more layers (Config 3: 79.10%) yields moderate gains. The best results come from combining the baseline architecture with data augmentation (RandomCrop, RandomHorizontalFlip, ColorJitter) and unfreezing the last ResNet block (layer4), allowing the backbone to adapt to the target dataset while regularizing through augmentation. All implementation details and training logs are available in the accompanying Jupyter notebook.

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References

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[3] *Lecture 9: Transfer Learning and Fine-Tuning*. Deep Learning Course, University of Bergamo. Prof. Subhankar Roy.

[4] *Lecture 10: Sequence Modelling and Attention*. Deep Learning Course, University of Bergamo. Prof. Subhankar Roy.

[5] *Lecture 11: NLP and Transformers*. Deep Learning Course, University of Bergamo. Prof. Subhankar Roy.

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